



University College of Engineering Nagercoil

Case Study Report:14

SafeGuard Insurance – Policy & Claims Analytics

Industry: Insurance

Modules: Reporting, Exploration, Dashboard, Story Making

Submitted by

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Question Scenario:

Company Overview:

SafeGuard Insurance is a well-established insurance provider offering **Health, Life, Motor, and Property insurance** across multiple regions. The company handles a large volume of policies and claims, making efficient claims processing and fraud detection critical to its operations.

Problem Statement:

The company is facing major operational challenges:

- Claim settlement delays beyond the IRDAI mandated 30 days
- Higher-than-average claim rejection rates
- Increasing number of fraudulent claims
- Lack of clear insights into agent performance and claim trends

Objective:

The Chief Actuary and Operations team aim to use IBM Cognos Analytics to:

- Monitor claim settlement performance
- Detect fraudulent claims
- Analyze agent productivity
- Improve customer satisfaction and compliance

Solution Approach:

1. Claims Settlement Reporting (Q1)

Grouping: Data grouped by Policy Type

Metrics:

- Total Claims
- Settled Claims
- Pending Claims
- Average Days to Settle

- Rejection Rate %

Conditional Styling:

- Red → Avg Days > 30 (IRDAI breach)
- Amber → 20–30
- Green → < 20

Prompts:

- Agent Branch
- Policy Type

2. Fraud Indicator Report (Q2)

Logic:

- Filter claims where Fraud Indicator = Yes

Calculations:

- Total Fraud Cases
- Total Fraud Amount

Conditional Styling:

- Claim Amount > ₹5,00,000 → Red Bold

3. Claims Trend Analysis (Q3)

Visualization:

- Report Visualization Prompt (Bar / Column)

Metrics:

- Monthly Claims Volume
- Settlement Rate

Filters:

- Policy Type
- Date Range

Insight:

- Identify months where settlement rate $< 70\%$

4. Data Exploration & Insights (Q4)

Analysis:

- Age Group vs Claim Frequency
- City vs Fraud Cases
- Agent Performance vs Rejection Rate
- Policy Type vs Settlement Time

5. Insurance Operations Dashboard (Q5)

KPI Tiles:

- Total Active Policies
- Claims This Month
- Average Settlement Days
- Fraud Cases Flagged

Visualizations:

- Bar → Claims by Policy Type
- Line → Monthly Claims Trend
- Pie → Claim Status Distribution
- Table → Agent Performance

6. Story for Regulatory Review (Q6)

Scenes:

- Policy Overview
- Claims Performance
- Fraud Analysis
- Agent Performance
- Customer Satisfaction
- Action Plan

Expected Outcome:

- Faster claim settlement
- Reduced fraud cases
- Improved regulatory compliance
- Better agent performance

- Increased customer satisfaction

Q1) [Reporting] Create a Claims Settlement Report grouped by Policy Type showing Total Claims, Settled Claims, Pending Claims, Average Days to Settle, and Rejection Rate %. Apply Conditional Styling: Avg Days > 30 in Red (IRDAI breach), 20–30 in Amber, < 20 in Green. Add prompts for Agent Branch and Policy Type. Header must include SafeGuard branding; Footer must include 'Regulatory Document'. Use Arial font, size 11.

Objective

Create a report grouped by Policy Type showing claims performance and settlement efficiency.

Step 1: Understanding the Requirement

The goal is to analyze:

- Claim volume
- Settlement performance
- Rejection trends

Step 2: Preparing the Data

Fields required:

- Claim ID
- Policy Type
- Claim Status
- Days to Settle
- Agent Branch

Step 3: Creating the Report

- Open Cognos → New Report
- Select List Report
- Add dataset

Step 4: Grouping the Data

- Drag Policy Type
- Apply grouping

Step 5: Creating Calculations

Total Claims:

```
count([Claim ID])
```

Settled Claims:

```
count(case when [Claim Status] = 'Settled' then 1 else null end)
```

Pending Claims:

```
count(case when [Claim Status] = 'Pending' then 1 else null end)
```

Average Days:

```
avg([Days to Settle])
```

Rejection Rate:

```
(count(case when [Claim Status] = 'Rejected' then 1 else null end) / count([Claim ID])) * 100
```

Step 6: Conditional Styling

- Red → >30 days
- Amber → 20–30
- Green → <20

10 Apr 2026 SafeGuard Insurance - Policy & Claims Analytics						
Policy Type	Total_Claims	Settled_Claims	Pending_Claim	Rejection Rate	Agent Branch	Days to Settle
Health	1	FALSE	FALSE	14,900	Hyderabad Branch	44
	1	FALSE	TRUE	14,900	Hyderabad Branch	3
	1	FALSE	FALSE	14,900	Delhi Branch	26
	1	FALSE	FALSE	14,900	Bengaluru Branch	30
	1	FALSE	FALSE	14,900	Chennai Branch	3
	1	FALSE	TRUE	14,900	Chennai Branch	30
	1	FALSE	FALSE	14,900	Mumbai Branch	36
	1	FALSE	FALSE	14,900	Chennai Branch	36

Step 7: Prompts

- Agent Branch
- Policy Type

Step 8: Header & Footer

Header: SafeGuard Insurance

Footer: Regulatory Document

					Branch	
1	FALSE	TRUE	14,900	Hyderabad Branch	52	
1	FALSE	TRUE	14,900	Chennai Branch	38	
1	FALSE	TRUE	14,900	Chennai Branch	49	
1	FALSE	TRUE	14,900	Delhi Branch	42	

Regulatory Document

Outcome:

10 Apr 2026 SafeGuard Insurance - Policy & Claims Analytics						
Policy Type	Total_Claims	Settled_Claims	Pending_Claim	Rejection Rate	Agent Branch	Days to Settle
Health	1	FALSE	FALSE	14,900	Hyderabad Branch	44
	1	FALSE	TRUE	14,900	Hyderabad Branch	3
	1	FALSE	FALSE	14,900	Delhi Branch	26
	1	FALSE	FALSE	14,900	Bengaluru Branch	30
	1	FALSE	FALSE	14,900	Chennai Branch	3
	1	FALSE	TRUE	14,900	Chennai Branch	30
	1	FALSE	FALSE	14,900	Mumbai Branch	36
	1	FALSE	FALSE	14,900	Chennai Branch	56
	1	FALSE	FALSE	14,900	Chennai Branch	23
	1	FALSE	FALSE	14,900	Chennai Branch	34
	1	FALSE	TRUE	14,900	Chennai Branch	3
	1	FALSE	TRUE	14,900	Chennai Branch	3
	1	FALSE	TRUE	14,900	Mumbai Branch	38
	1	FALSE	TRUE	14,900	Hyderabad Branch	58
	1	FALSE	TRUE	14,900	Bengaluru Branch	3
	1	FALSE	FALSE	14,900	Mumbai Branch	46
	1	FALSE	FALSE	14,900	Hyderabad Branch	33
	1	FALSE	TRUE	14,900	Bengaluru Branch	33
	1	FALSE	TRUE	14,900	Hyderabad Branch	3
	1	FALSE	FALSE	14,900	Bengaluru Branch	38
	1	FALSE	TRUE	14,900	Hyderabad Branch	52
	1	FALSE	TRUE	14,900	Chennai Branch	38
	1	FALSE	TRUE	14,900	Chennai Branch	49
	1	FALSE	TRUE	14,900	Delhi Branch	42

Regulatory Document

Conclusion

Provides visibility into delays and helps improve settlement performance.

Q2) [Reporting] Build a Fraud Indicator Report listing all claims where Fraud Indicator = Yes, showing Claim ID, Customer City, Claim Amount, Claim Type, and Rejection Reason. Apply Conditional Styling: Claim Amounts > ₹5,00,000 flagged in Red Bold. Add a summary count and total suspicious claim value at the footer. Sort by Claim Amount descending.

Objective

Identify and analyze fraudulent claims.

Step 1: Data Preparation

Fields:

- Claim ID
- City
- Claim Amount
- Claim Type
- Fraud Indicator
- Rejection Reason

Step 2: Report Creation

- Filter → Fraud Indicator = Yes

Step 3: Calculations

Total Fraud Cases:

```
count([Claim ID])
```

Total Fraud Amount:

```
sum([Claim Amount])
```

Step 4: Conditional Styling

- ₹5,00,000 → Red Bold

Step 5: Sorting

- Descending by Claim Amount

Outcome:

Zoom

Claim Type	Claim ID	City	Claim Amount	Rejection Reason
Theft	CLM00044	Bengaluru	990299	NA
Property Damage	CLM00191	Delhi	941148	NA
Property Damage	CLM00035	Chennai	844125	NA
Death Benefit	CLM00005	Hyderabad	830713	Incomplete Documents
Hospitalization	CLM00082	Pune	820085	NA
Natural Disaster	CLM00110	Jaipur	801152	NA
Natural Disaster	CLM00184	Mumbai	779343	NA
Hospitalization	CLM00102	Kolkata	757981	NA
Death Benefit	CLM00057	Pune	752909	NA
Death Benefit	CLM00194	Chennai	746724	NA
Natural Disaster	CLM00076	Chennai	721432	Fraud Suspected
Property Damage	CLM00143	Jaipur	704094	Exclusion Clause
Accident	CLM00018	Delhi	685535	NA
Death Benefit	CLM00192	Pune	563112	NA
Hospitalization	CLM00090	Bengaluru	543886	NA
Natural Disaster	CLM00019	Chennai	538877	Policy Lapsed
Natural Disaster	CLM00002	Delhi	517877	Fraud Suspected
Theft	CLM00159	Mumbai	503306	Exclusion Clause
Theft	CLM00068	Jaipur	499156	NA
Theft	CLM00027	Pune	464542	Duplicate Claim
Property Damage	CLM00181	Delhi	445951	NA
Property Damage	CLM00052	Kolkata	408645	NA
Accident	CLM00130	Chennai	392124	Duplicate Claim
Natural Disaster	CLM00032	Kolkata	332364	Incomplete Documents
Death Benefit	CLM00096	Jaipur	329022	Policy Lapsed
Theft	CLM00121	Jaipur	112697	NA
Accident	CLM00180	Mumbai	61845	NA
Death Benefit	CLM00020	Kolkata	61200	NA
Theft	CLM00064	Pune	5626	Policy Lapsed
Overall - Count	37			
Overall - Total			₹17,546,033.00	

Conclusion

Helps reduce financial loss by detecting suspicious claims.

Q3) [Reporting] Using Report Visualize Prompt, allow Regional Managers to select chart type (Bar or Column) to view Monthly Claims Volume vs Settlement Rate. Add prompts for Policy Type and Date Range. Summarize months where settlement rate dropped below 70% using Conditional Styling in Amber. Add a text annotation of the top 3 settlement

bottlenecks in the report.

Steps

Step 1: Understanding the Requirement

The goal is to create an interactive report where managers can:

- View Monthly Claims Volume
- Analyze Settlement Rate trends
- Switch between different chart types

Step 2: Preparing the Data

Ensure dataset contains:

- Claim Date (Month)
- Claim ID
- Claim Status
- Policy Type

Step 3: Creating the Report

- Open IBM Cognos Analytics
- Click **New** → **Report**
- Select **Chart Report**
- Add SafeGuard dataset

Step 4: Adding Visualization Prompt

- Drag **Report Visualization Prompt** from toolbox
- Configure options:
 - Bar Chart
 - Column Chart
- Link the prompt to the chart

Step 5: Building the Chart

- X-axis → Claim Date (Month)
- Y-axis → Count of Claim ID (Monthly Claims Volume)
- Add Settlement Rate as secondary measure

Settlement Rate:

```

    (count(case when [Claim Status] = 'Settled' then 1 else null
end)
/ count([Claim ID])) * 100

```



Step 6: Adding Filters

- Add **Policy Type** filter
- Add **Date Range** filter

This allows dynamic analysis based on user input

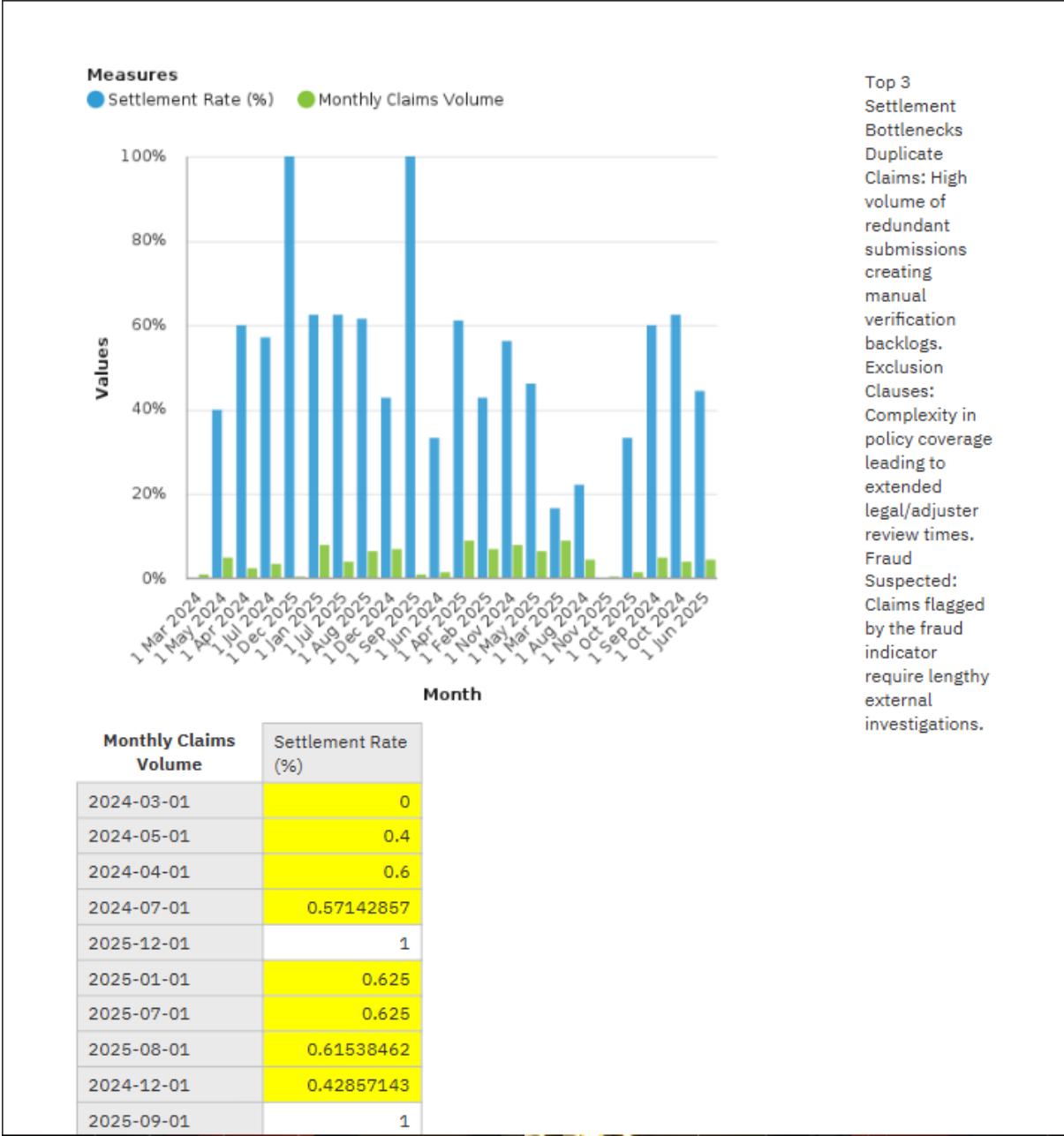
Step 7: Applying Conditional Styling

- Settlement Rate < 70% → Amber

This highlights performance issues

Step 8: Adding Insights

- Identify peak claim months
- Highlight months with low settlement rate



Monthly Claims Volume	Settlement Rate (%)
2024-03-01	0
2024-05-01	0.4
2024-04-01	0.6
2024-07-01	0.57142857
2025-12-01	1
2025-01-01	0.625
2025-07-01	0.625
2025-08-01	0.61538462
2024-12-01	0.42857143
2025-09-01	1

Monthly Claims Volume	Settlement Rate (%)
2025-05-01	0.46153846
2025-03-01	0.16666667
2024-08-01	0.22222222
2025-11-01	0
2025-10-01	0.33333333
2024-09-01	0.6
2024-10-01	0.625
2025-06-01	0.44444444

Top 3 Settlement Bottlenecks
Duplicate Claims: High volume of redundant submissions creating manual verification backlogs.
Exclusion Clauses: Complexity in policy coverage leading to extended legal/adjuster review times.
Fraud Suspected: Claims flagged by the fraud indicator require lengthy external investigations.

Step 9: Final Output Verification

- Check chart switching works
- Verify filters update data
- Ensure calculations are correct

Conclusion

Provides a dynamic view of claims trends and helps identify performance gaps.

Q4) [Exploration] Use Cognos Exploration to analyze SafeGuard's claims dataset. Explore whether Age Group or City correlates with higher claim frequency and fraud flags. Investigate the relationship between Agent Performance Rating and Claim Rejection Rate.

Identify which Policy Types have the longest average settlement times and what drives the delays

Steps

Step 1: Understanding the Requirement

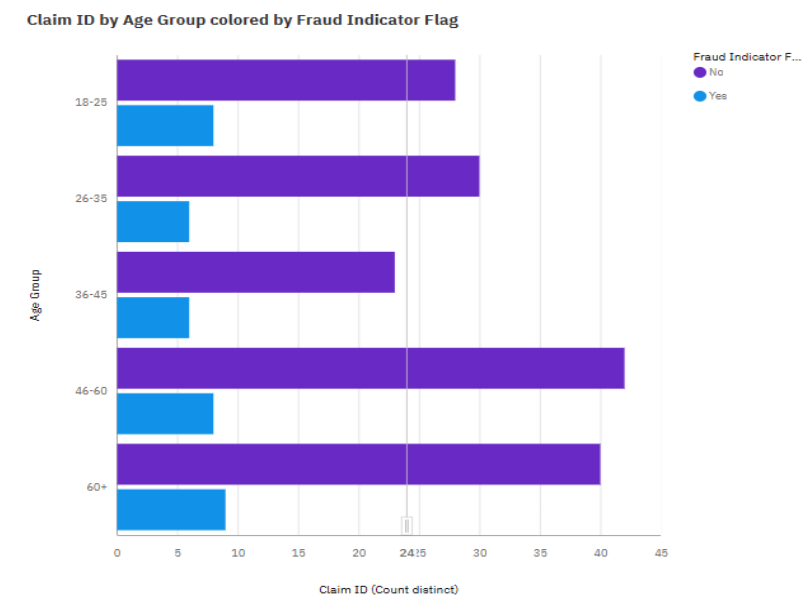
Exploration is used to discover hidden patterns and relationships in claims data rather than fixed reporting.

Step 2: Starting Exploration

- Open IBM Cognos Analytics
- Click **New** → **Exploration**
- Select SafeGuard dataset

Step 3: Age Group vs Claim Frequency

- Drag **Age Group**
- Add **Claim ID (Count)**
- Use **Bar Chart**



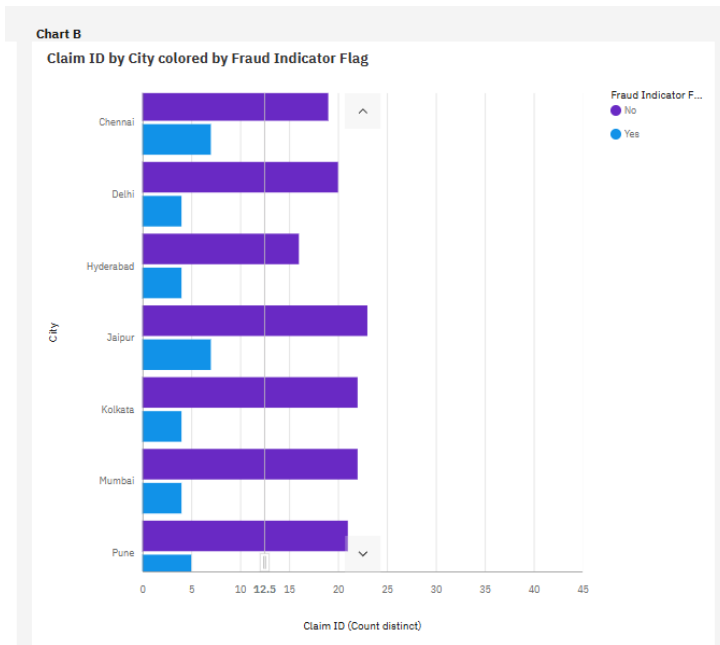
This shows which age groups file more claims

Step 4: City vs Fraud Analysis

- Drag **City**
- Create calculation:

```
case when [Fraud Indicator] = 'Yes' then 1 else 0 end
```

- Aggregate using **Sum**
- Use **Bar Chart**



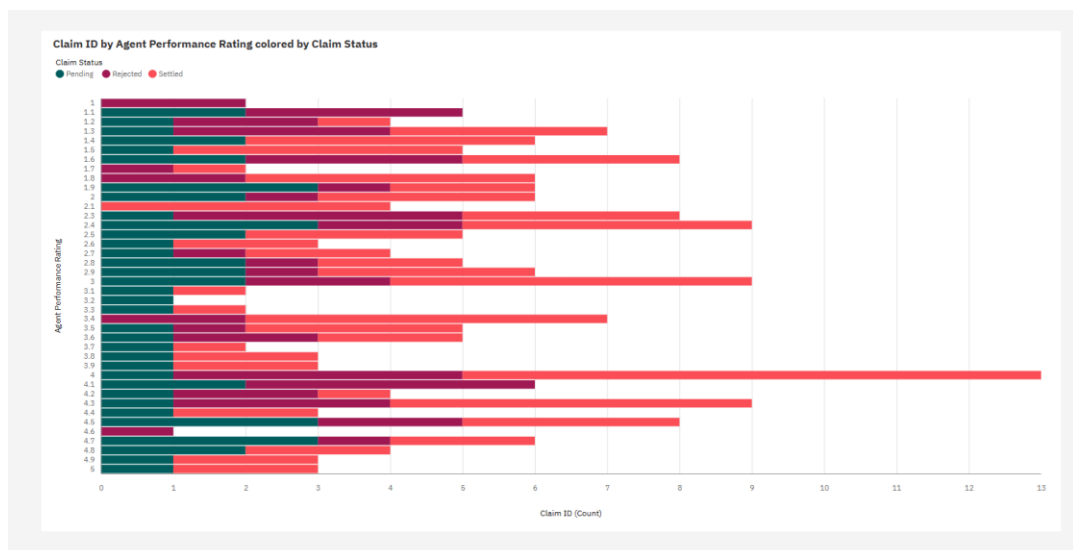
This identifies high-risk cities

Step 5: Agent Performance vs Rejection Rate

- Create calculation:

```
case when [Claim Status] = 'Rejected' then 1 else 0 end
```

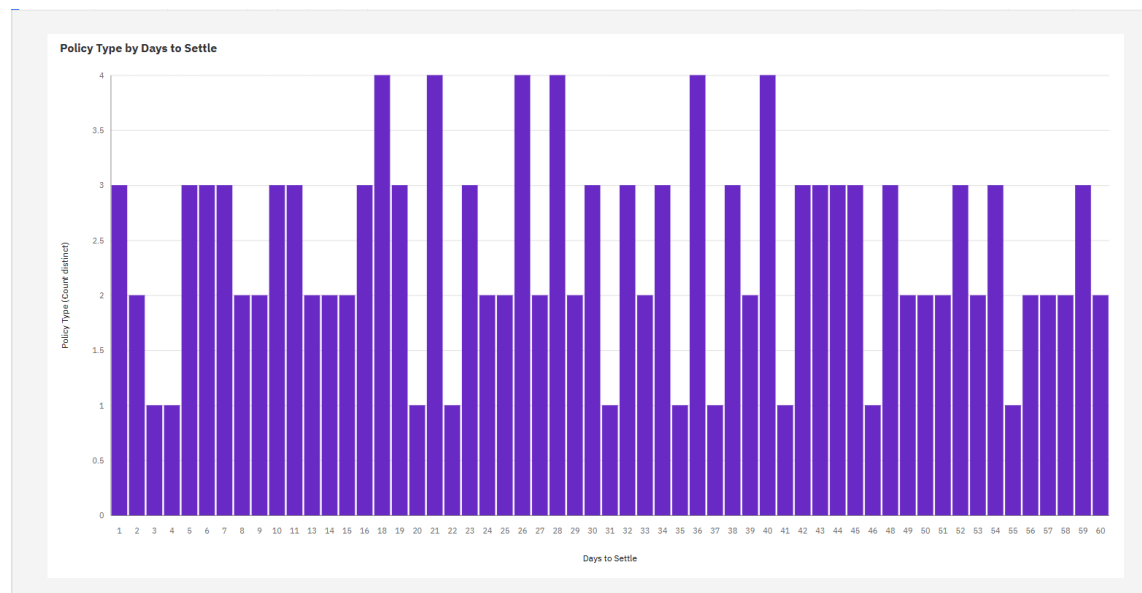
- Compare with **Agent Performance Rating**
- Use **Column Chart**



This shows impact of agent efficiency

Step 6: Policy Type vs Settlement Time

- Drag **Policy Type**
- Add **Avg(Days to Settle)**
- Use **Bar Chart**



This identifies delays by policy type

Step 7: Using AI Insights

- Click **Insights**
- Review auto-generated patterns
- Pin important visuals

Step 8: Final Review

- Validate charts
- Compare trends
- Ensure insights answer all objectives

Conclusion

Exploration helps identify root causes of fraud, delays, and inefficiencies.

Q5) [Dashboard] Design an Insurance Operations Dashboard with: (1) KPI tiles — Total Active Policies, Claims This Month, Average Settlement Days, Fraud Cases Flagged, (2) A Bar

chart for Claims by Policy Type, (3) A Line chart for Monthly Claims Trend, (4) A Pie chart for Claim Status Distribution (Settled/Pending/Rejected), (5) An Agent Performance table. Add Branch and Policy Type filters.

Steps

Step 1: Understanding the Requirement

The dashboard should provide a real-time overview of:

- Claims performance
- Settlement efficiency
- Fraud detection
- Agent performance

Step 2: Preparing the Data

Ensure dataset contains:

- Claim ID
- Policy Type
- Claim Status
- Days to Settle
- Fraud Indicator
- Agent Name
- Customer Satisfaction Score

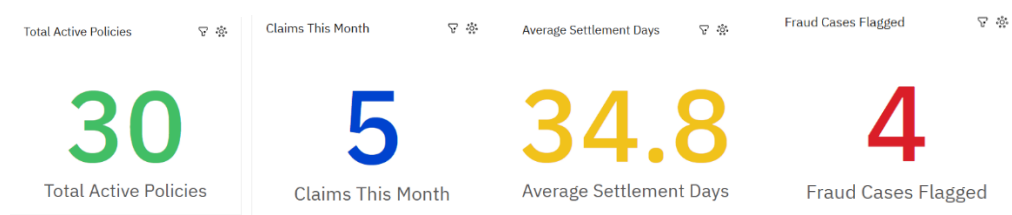
Step 3: Creating the Dashboard

- Open IBM Cognos Analytics
- Click **New** → **Dashboard**
- Choose layout (Grid / Freeform)
- Add dataset

Step 4: Creating KPI Tiles

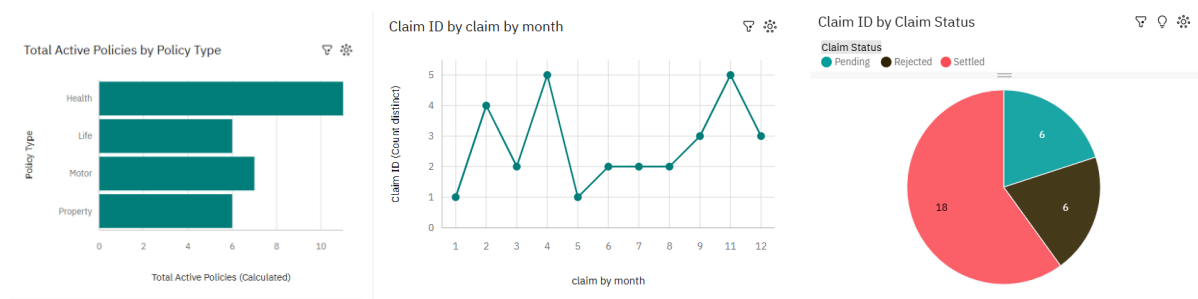
Add summary widgets:

- Total Active Policies
- Claims This Month
- Avg Settlement Days
- Fraud Cases Flagged



Step 5: Creating Visualizations

- **Bar Chart:** Claims by Policy Type
- **Line Chart:** Monthly Claims Trend
- **Pie Chart:** Claim Status Distribution
- **Table:** Agent Performance



Agent Name	Avg Performance Rating	Avg Performance Rating	Avg Satisfaction Score
Agent_Ananya	2.79	2.79	2.84
Agent_Deepak	3.04	3.04	2.75
Agent_Kiran	2.93	2.93	3.19
Agent_Priya	3.33	3.33	3.12
Agent_Ramesh	2.87	2.87	3.25
Agent_Suresh	2.9	2.9	2.78
Summary	2.79	2.79	2.75

Step 6: Adding Filters

- Add **Branch** filter
- Add **Policy Type** filter

Agent Branch

Select value

Bengaluru Branch

Chennai Branch

Delhi Branch

Hyderabad Branch

Mumbai Branch

Policy Type

Select value

Health

Life

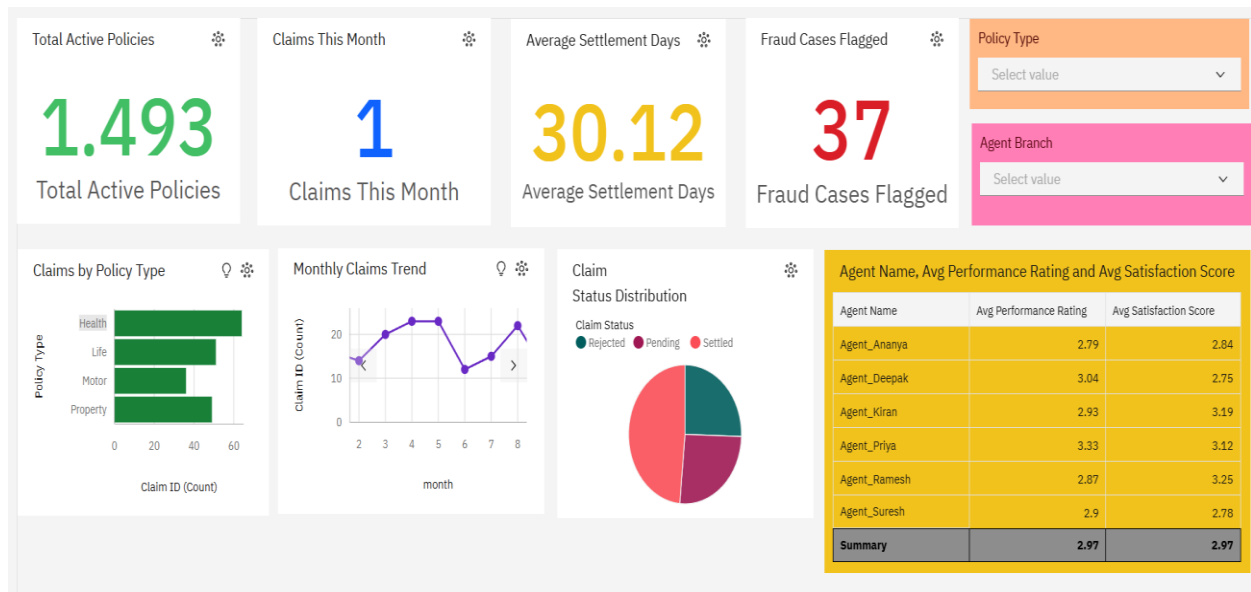
Motor

Property

These filters control all visuals

Step 7: Formatting the Dashboard

- Place KPIs at the top
- Arrange charts in grid layout
- Add title: *Insurance Operations Dashboard*



Step 8: Final Output Verification

- Check KPI values
- Verify charts display correctly
- Ensure filters work across all visuals

HTML Code:

This webpage is a professional insurance analytics dashboard called "SafeGuard Insurance Operations Hub". It serves as a centralized analytics portal that visualizes key insurance business metrics such as claims performance, policy distribution, fraud indicators, and agent productivity. The site is designed to integrate raw data from IBM Cognos Analytics into a clean and modern executive interface for real-time monitoring and decision-making.

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0, user-scalable=no">
  <title>Insurance Analytics | Cognos Live Dashboard</title>
  <!-- Google Fonts -->
  <link href="https://fonts.googleapis.com/css2?family=Inter:opsz,wght@14..32,300;400;500;600;700&display=swap"
rel="stylesheet">
  <!-- Font Awesome -->
  <link rel="stylesheet" href="https://cdn.jsdelivr.net/npm/font-awesome@6.0.0-beta3/css/all.min.css">
  <!-- Chart.js -->
  <script src="https://cdn.jsdelivr.net/npm/chart.js@4.4.0/dist/chart.umd.min.js"></script>
<style>
  * {
    margin: 0;
    padding: 0;
    box-sizing: border-box;
  }

  body {
    font-family: 'Inter', sans-serif;
    background: #f0f2f5;
    height: 100vh;
    overflow: hidden;
    color: #1e293b;
  }

  .login-container {
    display: flex;
    align-items: center;
    justify-content: center;
    min-height: 100vh;
    background: linear-gradient(135deg, #0f2b3d 0%, #1b4f72 100%);
  }

  .login-card {
    background: white;
    border-radius: 32px;
    box-shadow: 0 25px 50px -12px rgba(0, 0, 0, 0.25);
    width: 100%;
    max-width: 460px;
    padding: 2.5rem 2rem;
  }

  .login-header {
    text-align: center;
    margin-bottom: 2rem;
  }
```

```
}

.login-header .logo-icon {
  font-size: 3rem;
  color: #1e5f8e;
  background: #eef2ff;
  width: 70px;
  height: 70px;
  line-height: 70px;
  border-radius: 60px;
  display: inline-block;
  margin-bottom: 1rem;
}

.login-header h1 {
  font-size: 1.8rem;
  font-weight: 600;
  color: #0f2b3d;
}

.input-group {
  margin-bottom: 1.5rem;
}

.input-group label {
  display: block;
  font-weight: 500;
  font-size: 0.85rem;
  margin-bottom: 0.5rem;
  color: #334155;
}

.input-field {
  width: 100%;
  padding: 0.9rem 1rem;
  border: 1px solid #cbd5e1;
  border-radius: 18px;
  font-size: 1rem;
  font-family: 'Inter', sans-serif;
  transition: all 0.2s;
}

.input-field:focus {
  outline: none;
  border-color: #1e5f8e;
  box-shadow: 0 0 3px rgba(30, 95, 142, 0.15);
}
```

```
.login-btn {
  background: #1e5f8e;
  color: white;
  border: none;
  width: 100%;
  padding: 0.9rem;
  font-size: 1rem;
  font-weight: 600;
  border-radius: 40px;
  cursor: pointer;
}

.login-btn:hover {
  background: #124263;
}

.error-msg {
  background: #fee2e2;
  color: #b91c1c;
  padding: 0.7rem;
  border-radius: 14px;
  font-size: 0.8rem;
  text-align: center;
  margin-top: 1rem;
  display: none;
}

.dashboard-container {
  display: none;
  flex-direction: column;
  height: 100vh;
  width: 100%;
  overflow: hidden;
}

.app-header {
  background: white;
  box-shadow: 0 1px 3px rgba(0, 0, 0, 0.05);
  padding: 0 1.5rem;
  display: flex;
  align-items: center;
  justify-content: space-between;
  height: 64px;
  border-bottom: 1px solid #e2e8f0;
  flex-shrink: 0;
}

.logo-area {
```

```
display: flex;
align-items: center;
gap: 12px;
}

.logo-area i {
font-size: 28px;
color: #1e5f8e;
}

.logo-area h2 {
font-weight: 600;
font-size: 1.2rem;
color: #0f2b3d;
}

.user-info {
display: flex;
align-items: center;
gap: 20px;
}

.logout-btn {
background: transparent;
border: 1px solid #cbd5e1;
padding: 8px 16px;
border-radius: 30px;
cursor: pointer;
}

.main-content {
flex: 1;
overflow-y: auto;
padding: 1.5rem;
}

.kpi-row {
display: grid;
grid-template-columns: repeat(4, 1fr);
gap: 1.5rem;
margin-bottom: 1.5rem;
}

.kpi-card {
background: white;
border-radius: 20px;
padding: 1.25rem;
box-shadow: 0 1px 3px rgba(0,0,0,0.05);
```

```
border: 1px solid #e9eef3;
transition: transform 0.2s;
}

.kpi-card:hover {
  transform: translateY(-2px);
  box-shadow: 0 8px 25px rgba(0,0,0,0.08);
}

.kpi-title {
  font-size: 0.85rem;
  font-weight: 500;
  color: #64748b;
  text-transform: uppercase;
  letter-spacing: 0.5px;
  margin-bottom: 0.5rem;
}

.kpi-value {
  font-size: 2.5rem;
  font-weight: 700;
  color: #0f2b3d;
  line-height: 1.2;
}

.two-columns {
  display: grid;
  grid-template-columns: 1fr 1fr;
  gap: 1.5rem;
  margin-bottom: 1.5rem;
}

.chart-card {
  background: white;
  border-radius: 20px;
  padding: 1.25rem;
  border: 1px solid #e9eef3;
  box-shadow: 0 1px 3px rgba(0,0,0,0.05);
}

.chart-title {
  font-size: 1rem;
  font-weight: 600;
  color: #1e293b;
  margin-bottom: 1rem;
  padding-bottom: 0.5rem;
  border-bottom: 2px solid #eef2ff;
}
```

```
canvas {
  max-height: 250px;
  width: 100%;
}

.table-card {
  background: white;
  border-radius: 20px;
  padding: 1.25rem;
  border: 1px solid #e9eef3;
}

.data-table {
  width: 100%;
  border-collapse: collapse;
}

.data-table th {
  text-align: left;
  padding: 0.75rem 0.5rem;
  background: #f8fafc;
  font-weight: 600;
  font-size: 0.8rem;
  color: #475569;
}

.data-table td {
  padding: 0.7rem 0.5rem;
  border-bottom: 1px solid #eef2f6;
  font-size: 0.85rem;
}

.cognos-section {
  background: white;
  border-radius: 20px;
  padding: 1.25rem;
  border: 1px solid #e9eef3;
  margin-top: 1rem;
}

.cognos-iframe {
  width: 100%;
  height: 500px;
  border: none;
  border-radius: 12px;
}
```

```

.filter-row {
  display: flex;
  gap: 1rem;
  margin-bottom: 1rem;
  flex-wrap: wrap;
}

.filter-select {
  padding: 0.5rem 1rem;
  border: 1px solid #cbd5e1;
  border-radius: 12px;
  font-size: 0.85rem;
  background: white;
}

@media (max-width: 1024px) {
  .kpi-row { grid-template-columns: repeat(2, 1fr); }
  .two-columns { grid-template-columns: 1fr; }
}
</style>
</head>
<body>

<!-- LOGIN PAGE - Fields will be pre-filled but NOT auto-submit -->
<div id="loginPage" class="login-container">
  <div class="login-card">
    <div class="login-header">
      <div class="logo-icon"><i class="fas fa-chart-line"></i></div>
      <h1>Insurance Analytics</h1>
      <p>IBM Cognos Live + Real-time KPIs</p>
    </div>
    <form id="loginForm">
      <div class="input-group">
        <label><i class="far fa-envelope"></i> Username</label>
        <input type="text" id="username" class="input-field" placeholder="Enter username"
value="admin@insurance.com">
      </div>
      <div class="input-group">
        <label><i class="fas fa-lock"></i> Password</label>
        <input type="password" id="password" class="input-field" placeholder="Enter password" value="Welcome123">
      </div>
      <button type="submit" class="login-btn"><i class="fas fa-arrow-right-to-bracket"></i> Sign In</button>
      <div id="loginError" class="error-msg"></div>
    </form>
    <div style="margin-top: 1rem; text-align: center; font-size: 0.75rem; color: #6c7a91;">
      <i class="fas fa-check-circle"></i> Credentials are pre-filled – click Sign In to continue
    </div>
  </div>

```

```
</div>

<!-- DASHBOARD -->
<div id="dashboardPage" class="dashboard-container">
  <div class="app-header">
    <div class="logo-area">
      <i class="fas fa-chart-pie"></i>
      <h2>Insurance Analytics Hub</h2>
    </div>
    <div class="user-info">
      <span><i class="fas fa-user-circle"></i> <span id="displayUsername">User</span></span>
      <button id="logoutButton" class="logout-btn"><i class="fas fa-sign-out-alt"></i> Logout</button>
    </div>
  </div>

  <div class="main-content">
    <!-- KPI Cards -->
    <div class="kpi-row">
      <div class="kpi-card">
        <div class="kpi-title"><i class="fas fa-file-alt"></i> Total Active Policies</div>
        <div class="kpi-value" id="totalPolicies">1,493</div>
      </div>
      <div class="kpi-card">
        <div class="kpi-title"><i class="fas fa-calendar-alt"></i> Claims This Month</div>
        <div class="kpi-value" id="claimsThisMonth">1</div>
      </div>
      <div class="kpi-card">
        <div class="kpi-title"><i class="fas fa-clock"></i> Average Settlement Days</div>
        <div class="kpi-value" id="avgSettlement">30.12</div>
      </div>
      <div class="kpi-card">
        <div class="kpi-title"><i class="fas fa-flag"></i> Fraud Cases Flagged</div>
        <div class="kpi-value" id="fraudCases">37</div>
      </div>
    </div>

    <!-- Filters -->
    <div class="filter-row">
      <select id="policyTypeFilter" class="filter-select">
        <option value="all">Policy Type: All</option>
        <option value="Health">Health</option>
        <option value="Life">Life</option>
        <option value="Motor">Motor</option>
        <option value="Property">Property</option>
      </select>
      <select id="branchFilter" class="filter-select">
        <option value="all">Agent Branch: All</option>
        <option value="Downtown">Downtown</option>
      </select>
    </div>
  </div>
</div>
```

```
        <option value="Uptown">Uptown</option>
        <option value="Westside">Westside</option>
    </select>
    <button id="refreshDataBtn" class="filter-select" style="background:#1e5f8e; color:white; border:none;"><i
class="fas fa-sync-alt"></i> Refresh Data</button>
</div>

<!-- Charts -->
<div class="two-columns">
    <div class="chart-card">
        <div class="chart-title"><i class="fas fa-chart-bar"></i> Claims by Policy Type</div>
        <canvas id="claimsByPolicyChart"></canvas>
    </div>
    <div class="chart-card">
        <div class="chart-title"><i class="fas fa-chart-line"></i> Monthly Claims Trend</div>
        <canvas id="monthlyTrendChart"></canvas>
    </div>
</div>

<!-- Claim Status + Agent Table -->
<div class="two-columns">
    <div class="chart-card">
        <div class="chart-title"><i class="fas fa-chart-donut"></i> Claim Status</div>
        <canvas id="claimStatusChart"></canvas>
    </div>
    <div class="table-card">
        <div class="chart-title"><i class="fas fa-users"></i> Agent Performance</div>
        <table class="data-table">
            <thead>
                <tr><th>Agent Name</th><th>Avg Performance Rating</th><th>Avg Satisfaction Score</th></tr>
            </thead>
            <tbody id="agentTableBody"></tbody>
        </table>
    </div>
</div>

<!-- Summary -->
<div class="kpi-row" style="margin-bottom:1.5rem">
    <div class="kpi-card">
        <div class="kpi-title">Overall Performance Rating</div>
        <div class="kpi-value" id="overallRating">2.97</div>
    </div>
    <div class="kpi-card">
        <div class="kpi-title">Overall Satisfaction Score</div>
        <div class="kpi-value" id="overallSatisfaction">2.97</div>
    </div>
</div>
```

```

<!-- Live Cognos Dashboard -->
<div class="cognos-section">
  <div class="chart-title"><i class="fab fa-ibm"></i> IBM Cognos Live Analytics (Real-Time)</div>
  <iframe id="cognosIframe" class="cognos-iframe"
    src="https://ap2.ca.analytics.ibm.com/bi/?perspective=dashboard&pathRef=.my_folders%2FEXP%253A5&close
WindowOnLastView=true&ui_appbar=false&ui_navbar=false&shareMode=embedded&action=view&mode=dashboa
rd&subView=model0000019d8012a9ca_00000004&nav_filter=true"
    frameborder="0" allow="encrypted-media" allowfullscreen>
  </iframe>
</div>
</div>
</div>

<script>
  // Demo data from screenshot
  const demoData = {
    totalPolicies: 1493,
    claimsThisMonth: 1,
    avgSettlement: 30.12,
    fraudCases: 37,
    claimsByPolicy: { Health: 45, Life: 28, Motor: 62, Property: 38 },
    monthlyTrend: [4, 6, 5, 8, 7, 9, 12, 10],
    claimStatus: { Settled: 142, Pending: 35, Rejected: 18 },
    agents: [
      { name: "Agent_Ananya", perf: 2.79, sat: 2.84 },
      { name: "Agent_Deepak", perf: 3.04, sat: 2.75 },
      { name: "Agent_Kiran", perf: 2.93, sat: 3.19 },
      { name: "Agent_Priya", perf: 3.33, sat: 3.12 },
      { name: "Agent_Ramesh", perf: 2.87, sat: 3.25 },
      { name: "Agent_Suresh", perf: 2.90, sat: 2.78 }
    ],
    overallRating: 2.97,
    overallSatisfaction: 2.97
  };

  let charts = {};

  function renderKPIs() {
    document.getElementById('totalPolicies').innerText = demoData.totalPolicies.toLocaleString();
    document.getElementById('claimsThisMonth').innerText = demoData.claimsThisMonth;
    document.getElementById('avgSettlement').innerText = demoData.avgSettlement.toFixed(2);
    document.getElementById('fraudCases').innerText = demoData.fraudCases;
    document.getElementById('overallRating').innerText = demoData.overallRating.toFixed(2);
    document.getElementById('overallSatisfaction').innerText = demoData.overallSatisfaction.toFixed(2);
  }

  function renderAgentTable() {
    const tbody = document.getElementById('agentTableBody');

```

```

tbody.innerHTML = '';
demoData.agents.forEach(agent => {
  tbody.innerHTML +=
`<tr><td>${agent.name}</td><td>${agent.perf.toFixed(2)}</td><td>${agent.sat.toFixed(2)}</td></tr>`;
});
}

function renderCharts() {
  if (charts.claimsByPolicy) charts.claimsByPolicy.destroy();
  const ctx1 = document.getElementById('claimsByPolicyChart').getContext('2d');
  charts.claimsByPolicy = new Chart(ctx1, {
    type: 'bar',
    data: {
      labels: Object.keys(demoData.claimsByPolicy),
      datasets: [{ label: 'Claim ID (Count)', data: Object.values(demoData.claimsByPolicy), backgroundColor:
'#1e5f8e', borderRadius: 8 }]
    },
    options: { responsive: true, maintainAspectRatio: true }
  });

  if (charts.monthlyTrend) charts.monthlyTrend.destroy();
  const ctx2 = document.getElementById('monthlyTrendChart').getContext('2d');
  charts.monthlyTrend = new Chart(ctx2, {
    type: 'line',
    data: {
      labels: ['Jan', 'Feb', 'Mar', 'Apr', 'May', 'Jun', 'Jul', 'Aug'],
      datasets: [{ label: 'Claims', data: demoData.monthlyTrend, borderColor: '#e67e22', backgroundColor:
'rgba(230,126,34,0.1)', tension: 0.3, fill: true }]
    },
    options: { responsive: true }
  });

  if (charts.claimStatus) charts.claimStatus.destroy();
  const ctx3 = document.getElementById('claimStatusChart').getContext('2d');
  charts.claimStatus = new Chart(ctx3, {
    type: 'doughnut',
    data: {
      labels: Object.keys(demoData.claimStatus),
      datasets: [{ data: Object.values(demoData.claimStatus), backgroundColor: ['#2ecc71', '#f39c12', '#e74c3c'] }]
    }
  });
}

function applyFilters() {
  const policyFilter = document.getElementById('policyTypeFilter').value;
  if (policyFilter !== 'all') {
    let filteredClaims = { ...demoData.claimsByPolicy };
    Object.keys(filteredClaims).forEach(key => {

```

```

        if (key !== policyFilter) filteredClaims[key] = 0;
    });
    if (charts.claimsByPolicy) {
        charts.claimsByPolicy.data.datasets[0].data = Object.values(filteredClaims);
        charts.claimsByPolicy.update();
    }
} else {
    if (charts.claimsByPolicy) {
        charts.claimsByPolicy.data.datasets[0].data = Object.values(demoData.claimsByPolicy);
        charts.claimsByPolicy.update();
    }
}
}

function refreshAllData() {
    demoData.claimsThisMonth = Math.floor(Math.random() * 10) + 1;
    demoData.fraudCases = Math.floor(Math.random() * 50) + 20;
    demoData.totalPolicies = 1493 + Math.floor(Math.random() * 50);
    renderKPIs();
    demoData.monthlyTrend = demoData.monthlyTrend.map(v => v + (Math.random() - 0.5) * 2);
    renderCharts();
}

// Authentication
function isAuthenticated() { return sessionStorage.getItem('cognos_auth') === 'true'; }
function setAuth(val, username) {
    sessionStorage.setItem('cognos_auth', val);
    if(username) sessionStorage.setItem('cognos_user', username);
}

// Login form submission (MANUAL - user must click Sign In)
document.getElementById('loginForm').addEventListener('submit', (e) => {
    e.preventDefault();
    const user = document.getElementById('username').value.trim();
    const pass = document.getElementById('password').value.trim();
    if(user && pass) {
        setAuth('true', user);
        document.getElementById('loginPage').style.display = 'none';
        document.getElementById('dashboardPage').style.display = 'flex';
        document.getElementById('displayUsername').innerText = user;
        renderKPIs();
        renderAgentTable();
        renderCharts();
    } else {
        document.getElementById('loginError').innerText = 'Please enter username & password';
        document.getElementById('loginError').style.display = 'block';
    }
});

```

```

document.getElementById('logoutButton').addEventListener('click', () => {
  sessionStorage.clear();
  document.getElementById('loginPage').style.display = 'flex';
  document.getElementById('dashboardPage').style.display = 'none';
  // Clear the error message if any
  document.getElementById('loginError').style.display = 'none';
});

document.getElementById('policyTypeFilter').addEventListener('change', applyFilters);
document.getElementById('branchFilter').addEventListener('change', applyFilters);
document.getElementById('refreshDataBtn').addEventListener('click', refreshAllData);

// If already logged in, show dashboard
if(isAuthenticated()) {
  document.getElementById('loginPage').style.display = 'none';
  document.getElementById('dashboardPage').style.display = 'flex';
  document.getElementById('displayUsername').innerText = sessionStorage.getItem('cognos_user') || 'User';
  renderKPIs();
  renderAgentTable();
  renderCharts();
}
</script>
</body>
</html>

```

Webpage Design:

The website follows a modular **2-page architecture** within a unified environment:

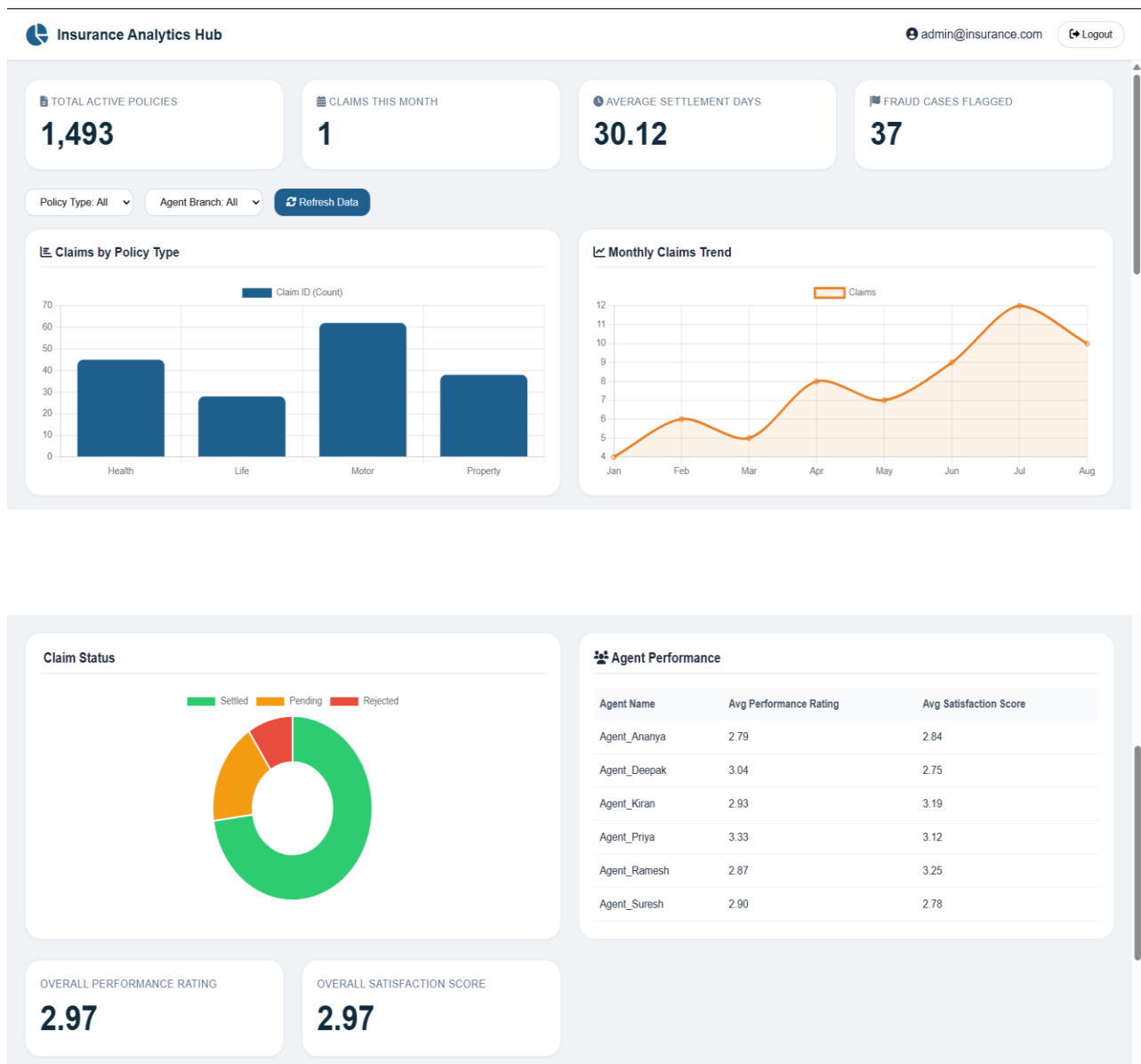
1. Login Page:

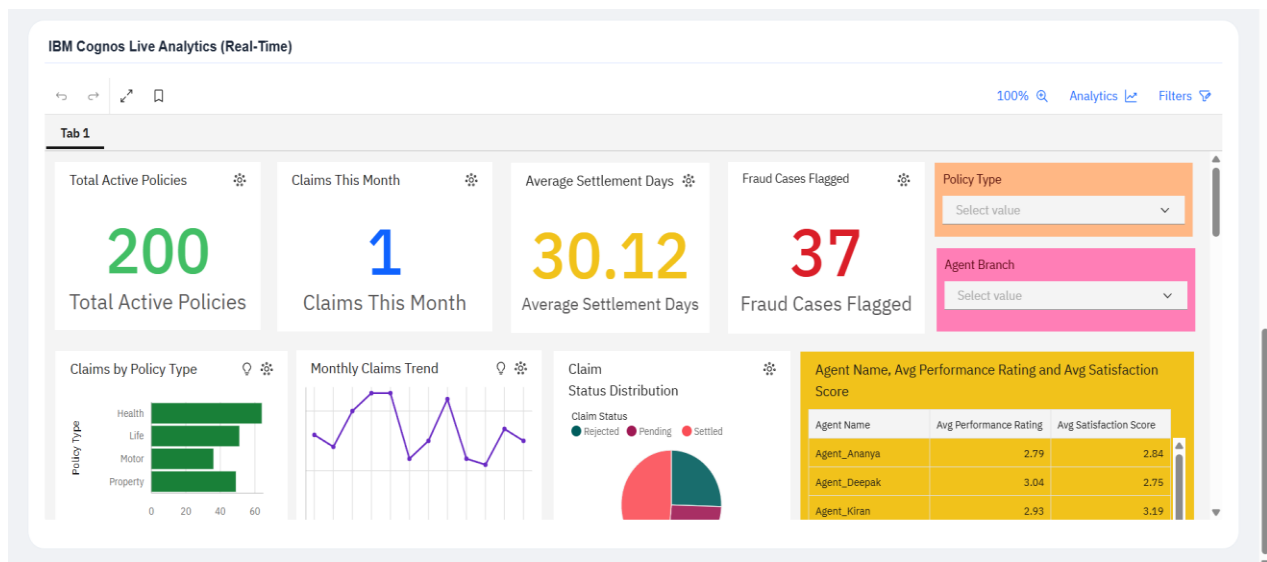
- A secure access portal for **insurance analysts, managers, and operations teams**
- Designed with **SafeGuard Insurance branding**
- Provides a simple and professional login interface with user authentication and logout functionality

2. Dashboard Page:

- **KPI Scorecards:**
Displays key insurance metrics such as:
 - Total Active Policies
 - Claims This Month
 - Average Settlement Days

- Fraud Cases Flagged
- **Analytics Visualizations:**
Interactive charts providing insights into:
 - **Claims by Policy Type** (Health, Life, Motor, Property)
 - **Monthly Claims Trend**
 - **Claim Status Distribution** (Settled, Pending, Rejected)
 - **Agent Performance Analysis** (Average Rating & Customer Satisfaction Score)
- **Cognos Live View:**
A dedicated section to embed the **IBM Cognos Analytics Dashboard**, enabling deeper analysis and real-time data exploration directly within the webpage





Conclusion

Dashboard provides a centralized and interactive view of insurance operations for better decision-making.

Q6) [Story Making] Create a Cognos Story for SafeGuard's Quarterly Regulatory & Operations Review: Scene 1 — Policy Portfolio & Renewal Overview, Scene 2 — Claims Volume & Settlement Performance, Scene 3 — Fraud Detection & Rejection Analysis, Scene 4 — Agent Productivity & Branch Performance, Scene 5 — Customer Satisfaction Trends, Scene 6 — Action Plan for IRDAI Compliance & Fraud Reduction.

Objective

Present insights through storytelling.

Steps

- Create Story
- Add scenes
- Add visuals
- Add insights

Scenes

- Introduction

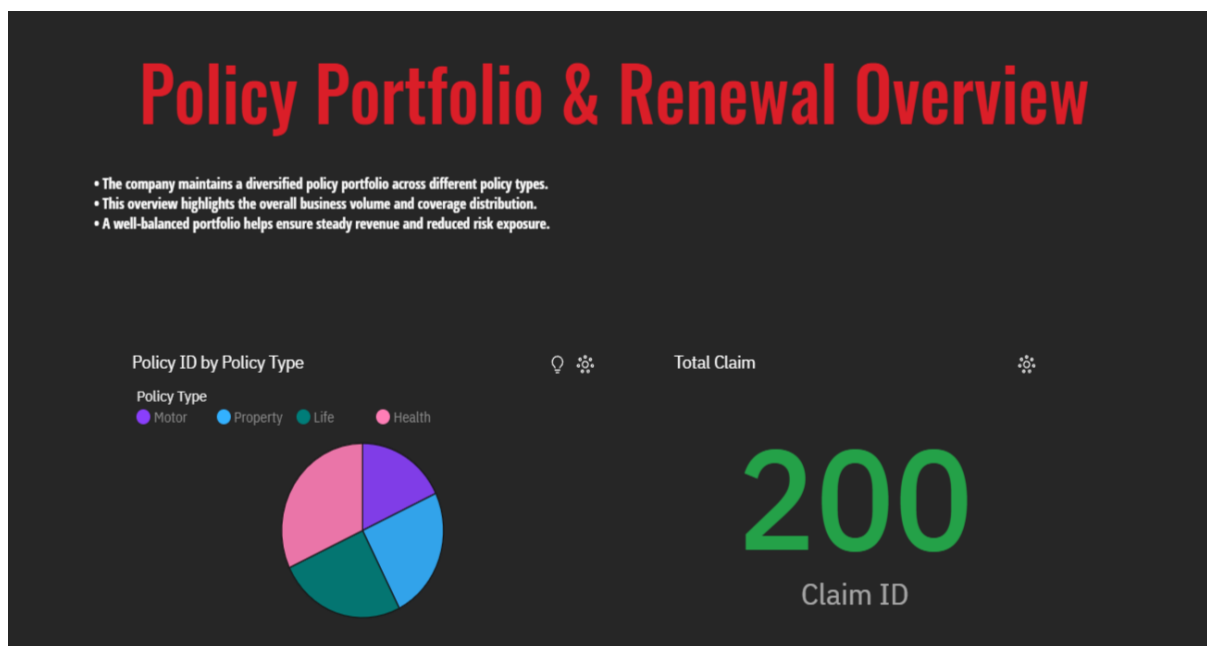
SafeGuard Insurance - Policy & Claims Analytics

- This analysis provides an overview of policy distribution and claim activities across the organization.
- It helps identify trends in claims, settlements, and overall business performance.
- The insights support better decision-making to improve efficiency and manage risk effectively.

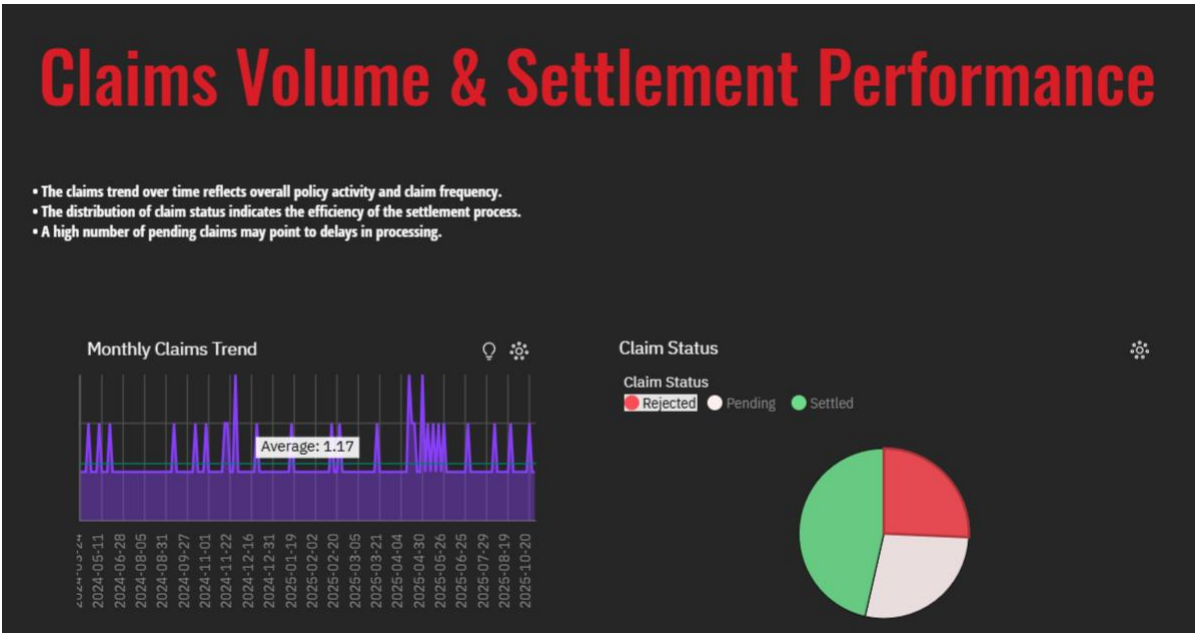


An illustration showing two hands interacting with a document titled 'INSURANCE POLICY'. One hand is holding the top left corner of the document, while the other hand is using a pen to sign the bottom right. The document has a blue circular stamp on the bottom left and several lines of placeholder text. The background is a solid blue color.

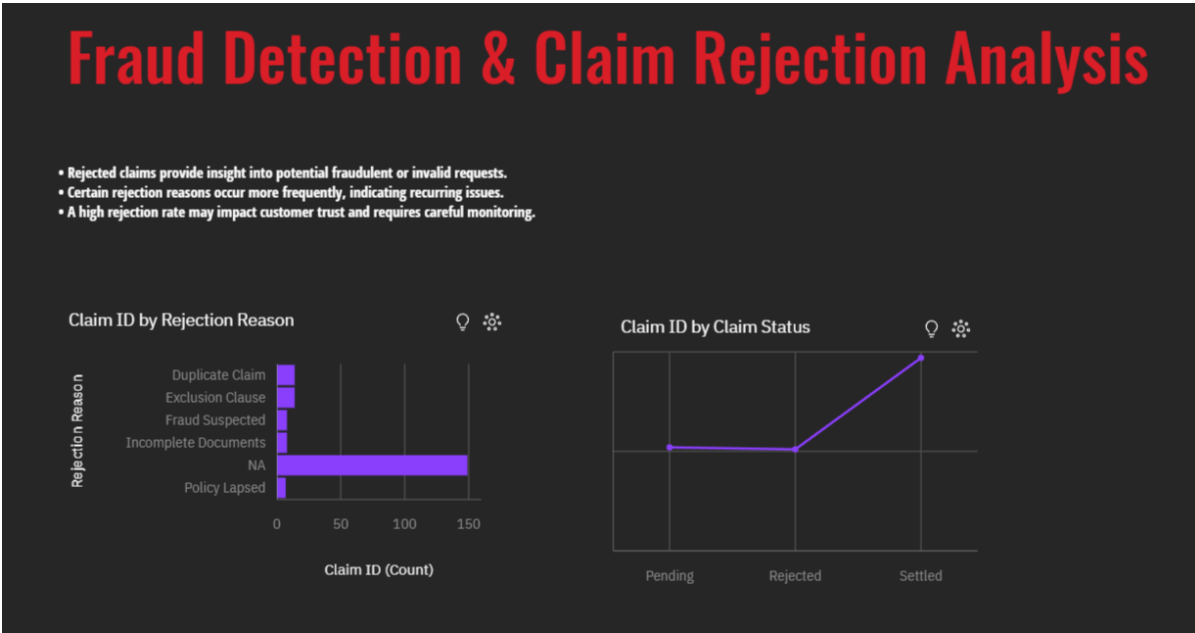
- Policy Overview



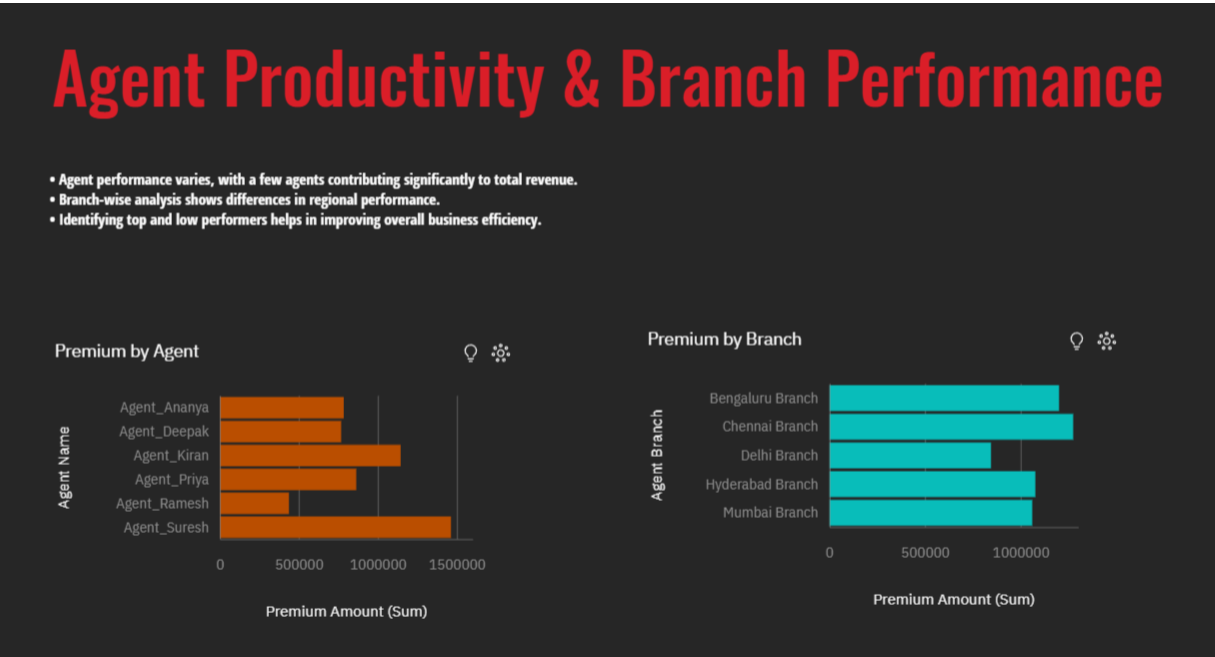
- Claims Performance



- Fraud Analysis



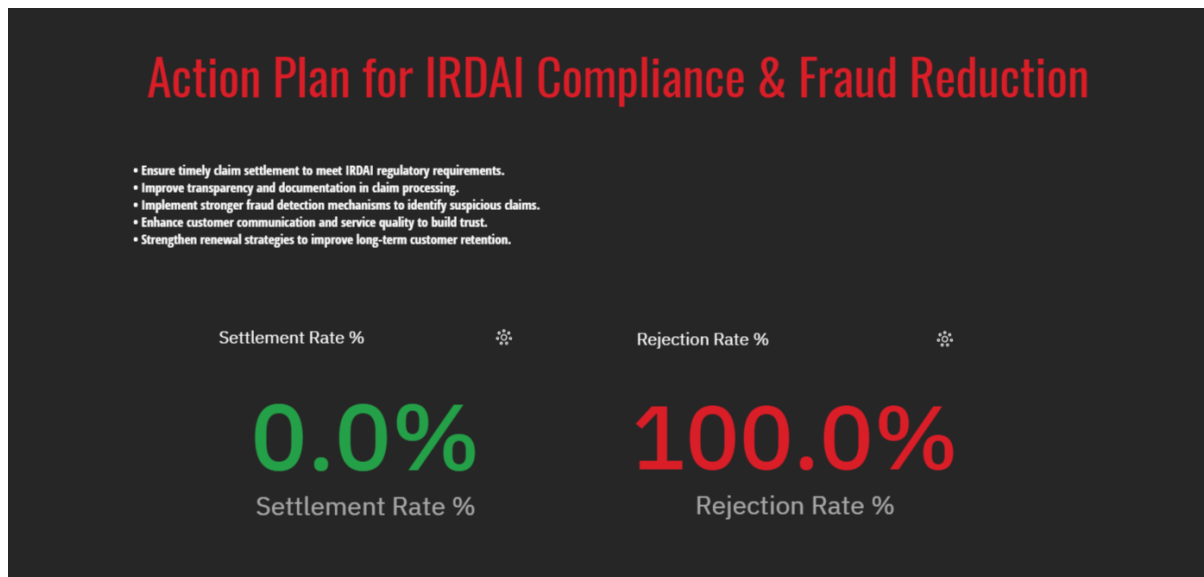
- Agent Performance



- Customer Satisfaction



- Action Plan



Conclusion

By following this 7-scene structure, the analysis moves from understanding current performance to recommending actionable improvements. This story offers SafeGuard Insurance management a clear roadmap to minimize settlement delays, strengthen fraud detection, and enhance agent productivity, leading to improved efficiency, higher customer satisfaction, and better compliance with IRDAI guidelines.

Project GitHub link: <https://sankaranarayanan-007.github.io/SafeGuard-Insurance---Policy-Claims-Analytics/>

GitHub Repository link: <https://github.com/SankaraNarayanan-007/SafeGuard-Insurance---Policy-Claims-Analytics.git>
